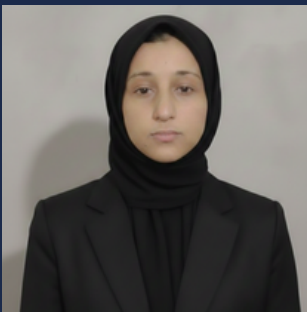




Kanwal Fatima

AI Student | Software Developer

CONTACT

EMAIL

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PHONE

+92 329-7808416

LOCATION

Islamabad, Pakistan

GITHUB

github.com/KanwalAi

LINKEDIN

<https://www.linkedin.com/in/kanwal-fatima-72a352357/>

SKILLS

LANGUAGES

Python, C++, PHP, JavaScript, SQL, C#, HTML/CSS, x86 Assembly
WEB

PHP, MySQL, Bootstrap, AJAX, Responsive Design

ROBOTICS

Arduino, ESP32, MQTT, IR/Ultrasonic Sensors

TOOLS

Git, GitHub, VS Code, XAMPP, Visual Studio

CONCEPTS

Object Oriented Programming, Data Structures, State Machines, Collision Detection

ACHIEVEMENT

PAI & IDS Robotics Exhibition
Participant (ISB FSC Certificate)

EDUCATION

B.S. Artificial Intelligence
FAST-NUCES, Islamabad
2024 – Present (Exp. 2028)

Kanwal Fatima

Artificial Intelligence Student | Software Developer | Web Development • Robotics • AI Projects

PROFILE

AI student at FAST-NUCES with hands-on experience in Full-Stack Web Development, Game Development, Robotics, and Systems Programming. Skilled in PHP, MySQL, Bootstrap, JavaScript, C++, Python, and Arduino. Passionate about solving real-world problems through software engineering and AI. Portfolio available on GitHub: github.com/KanwalAi

TECHNICAL PROJECTS

RainRover – Autonomous Vehicle

Arduino • Sensors • Servo • Motor Driver

- Arduino robot with IR line-following & self-parking.
- Rain sensor triggers automated convertible rooftop.
- Exhibited at PAI & IDS Robotics Exhibition (ISB FSC).

ElectroStore – E-Commerce Platform

PHP • MySQL • Bootstrap • JavaScript

- Full-stack e-commerce with customer & admin modules.
- Auth, product filtering, order management, reviews.
- Secure session login with DB-driven product displays.

Subway Surfers 2D – Infinite Runner

C++ • SFML • OOP

- 3-lane runner using inheritance, polymorphism & OOP.
- Dynamic obstacles, collision detection & power-ups.
- State machines for menu/gameplay/game-over flow.

Xonix – Territory Capture Game

C++ • SFML • Multiplayer • OOP

- Classic Xonix with multiplayer & dynamic difficulty.
- Power-ups, scoreboards & modular multi-file design.
- Encapsulation and reusable game components.

EXPERIENCE

Volunteer Student Tutor — Pehli Kiran School

Civics & Community Engagement Project

- Broke down complex subjects into digestible lessons for diverse learners.
- Customized teaching methods to support individual student learning paces.

OpticCLI – AI CLI Assistant

C# • .NET • AI APIs • Automation

- AI-powered command-line assistant developed in C# and .NET.
- Supports natural language commands, task automation, and workflow enhancement.
- Designed to improve developer productivity through AI-assisted execution.

Halal Delights – Food Ordering

PHP • MySQL • Bootstrap • PHPMailer

- Restaurant ordering platform with full CRUD for menus.
- PHPMailer for customer-admin email notifications.
- Cart management, admin dashboard, secure login.

Rush Hour Taxi – Assembly Game

x86 Assembly • MASM • Irvine32

- Arcade taxi simulation built entirely in x86 Assembly.
- Multiple modes, file-based leaderboards & sounds.
- Sorting algorithms for score management & save state.

Jan – June 2025